

A multi-criteria triage of genome-scale CD4+ T-cell Perturb-seq nominates druggable regulators of a pathogenic program, with BHLHE40 as a worked example

Abstract

Genome-scale perturbation atlases in primary human cells promise a rational route to therapeutic-target selection, but a single atlas returns thousands of statistically significant regulators and no principled way to choose among them. We built a reproducible, multi-criteria triage that scores every perturbation in a genome-scale CRISPRi Perturb-seq atlas of primary human CD4+ T cells (1) on four axes relevant to autoimmune-target selection: suppression of a pathogenic Th1/GM-CSF effector program, effect-selectivity for T cells, genetic support of the perturbation's regulon, and degradability of the encoded protein. Ranking all 1,857 gene perturbations that carry a cross-cell-type comparison, we show that the transcription factor **BHLHE40** is a joint standout (composite rank 3 of 1,857; top 0.2%) even though it is first on none of the four axes individually. As a worked example we then subject BHLHE40 to the scrutiny a lead nomination must survive. Its knockdown steers cells out of the pathogenic program only under stimulation (CSF2/GM-CSF $z = -4.8$, adj. $P = 8 \times 10^{-5}$ at 48 h; T-bet down; regulatory IL-10 up), and this effect is T-cell-selective at the level of both the perturbation contrast and the immune-weighted regulon. The knockdown signature reproduces the murine BHLHE40 cytokine axis (CSF2, IFN- γ , IL-10 concordant across species) but not the genome-wide wiring, and is confirmed in rheumatoid-arthritis patient CD4+ T cells, with partial support from an independent CRISPR-activation platform (a weak gain-of-function hit in an IL-2 bin; that screen did not read out GM-CSF or IL-10). A formal bootstrap mediation apports the indirect GM-CSF and IL-10 effects through the direct target T-bet (TBX21: 19% and 49% mediated). The triage is deliberately size-controlled on the genetic axis, so that broadly-acting regulators are not credited for large-regulon artifacts — under both a matched-null comparison and a formal MAGMA competitive gene-set test (run here against MS, RA and IBD; all null) BHLHE40's regulon is genetically ordinary at the set level, and its genetic support is asserted only at the member-gene level and for the two diseases (IBD, MS) its GM-CSF/Th1 axis predicts. The encoded protein is degrader-amenable but lacks a classical small-molecule pocket, and the downstream GM-CSF axis it controls has clinical precedent, repeated single-cytokine efficacy failures, and a defined on-target safety signal — arguments for intervening at the regulator rather than one effector. The triage, the ranked shortlist of 1,857 perturbations, and the BHLHE40 dossier are provided as a template for turning perturbation atlases into prioritized, falsifiable target hypotheses — not validated therapeutics.

Introduction

Autoimmune and inflammatory diseases are driven in part by CD4+ T cells that adopt a pathogenic effector state, and a recurring therapeutic goal is to disable that state without collapsing protective immunity. Human genetics repeatedly implicates T-cell regulatory circuitry: fine-mapped autoimmune risk variants concentrate in enhancers active in stimulated T cells (2), and the GM-CSF-producing, T-bet-high effector arm is a validated axis of pathogenicity in models of neuroinflammation (3, 4). Yet most transcription factors that shape this state have been considered undruggable, and the field has lacked an unbiased, human-primary map of which regulators actually control the pathogenic program.

Genome-scale CRISPRi Perturb-seq in primary human CD4+ T cells now supplies that map (1),

following the development of pooled CRISPR screening and single-cell perturbation readouts in T cells and other systems (12, 13, 14, 15). The difficulty is no longer identifying regulators but choosing among them: at a 5% false-discovery threshold the atlas contains hundreds of perturbations that significantly move a pro-inflammatory module, and statistical significance alone does not tell a wet-lab team which one to pursue. A useful nomination must clear several independent bars at once — it must move the disease-relevant program, do so with some selectivity for the target cell type, have genetic or orthogonal support, and correspond to a protein that can plausibly be drugged.

Here we make the *selection procedure* the contribution. We define a four-axis triage and apply it to every perturbation in the atlas, producing a genome-wide prioritization landscape and a reusable ranked shortlist. We then use the single highest-ranked novel regulator, BHLHE40, as a worked example: a stress-test of what the triage nominates, carried through cross-species comparison, independent human datasets, chromatin-state grounding, and an explicit accounting of where the evidence is strong and where it is not. BHLHE40 is a basic helix-loop-helix factor previously characterized almost entirely in mice, where it promotes GM-CSF and restrains IL-10 in pathogenic T cells (5, 6, 7, 8, 9); a smaller human study reports the same repression of IL-10 and IFN- γ (10). Whether it controls the analogous program in primary human CD4+ T cells, and whether it survives a target-selection stress-test, has not been tested at genome scale.

Results

A four-axis triage scores every perturbation in the atlas

We reasoned that a defensible autoimmune-target nomination must satisfy four criteria that are individually necessary and jointly rare. We therefore scored each perturbation on four axes, all computed at the 48-hour stimulation timepoint where the pathogenic program is most fully expressed. **Axis 1, program-suppression:** the signed effect of the knockdown on a 125-gene pathogenic Th1/GM-CSF effector module derived from the atlas, where a more negative value means stronger suppression. **Axis 2, effect-selectivity:** the discordance between the perturbation's effect in CD4+ T cells and in K562 cells (1 – Pearson r across shared downstream genes), a proxy for cell-type-specific wiring. **Axis 3, regulon-genetics:** the enrichment of the perturbation's own significant downstream gene set for autoimmune GWAS genes. **Axis 4, degradability:** a tractability class from structural and modality analysis, scored for the top composite candidates. Restricting to the 1,857 gene perturbations that carry a K562 comparison, we z-normalized the first three axes and ranked perturbations by their mean (Fig. 1).

The landscape shows that the axes are largely independent — the hardest suppressors (CBFB, BTAF1) are not the most selective, and the most selective perturbations (PCMT1, GPANK1) barely move the module — so joint performance is informative and rare. **BHLHE40 is a joint standout: composite rank 3 of 1,857 (top 0.2%; bootstrap 95% CI 1-6, top-1% in 100% of 2,000 resamples), at the 99.7th percentile for program-suppression, the 90.8th for effect-selectivity, and the 85.8th for regulon-genetics, yet first on none of the three.** It occupies the high-suppression, high-selectivity corner of the landscape (Fig. 1) while retaining top-quartile genetic support, and it is degrader-amenable (below). The full scored table for all 1,857 perturbations is provided as a reusable shortlist; the remainder of this paper uses BHLHE40 as a worked example to show what the triage nominates and how such a nomination holds up under scrutiny.

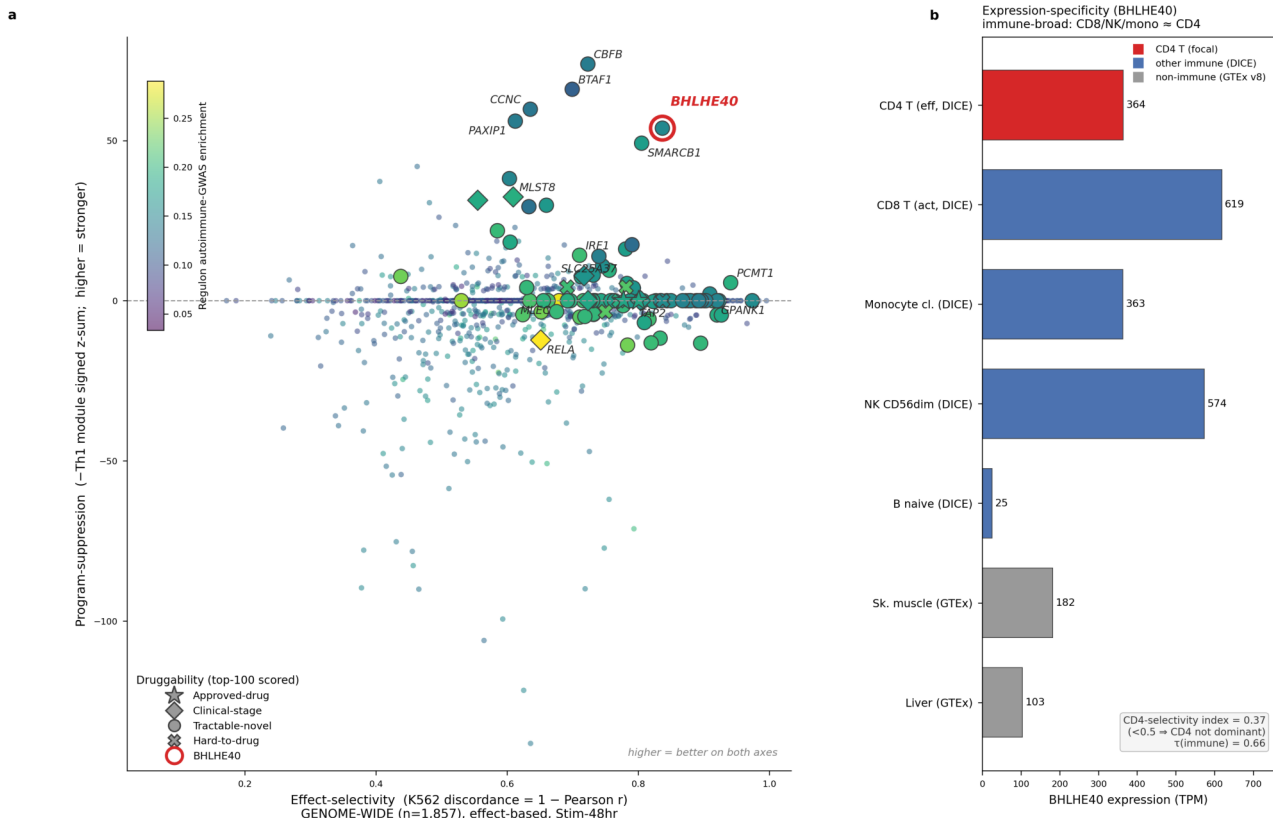


Fig. 1. Genome-wide triage identifies BHLHE40 as a joint standout, with multi-tissue selectivity. (A) Each point is one of 1,857 gene perturbations with a K562 comparison, at Stim-48hr. x , effect-selectivity (CD4-versus-K562 discordance, $1 - \text{Pearson } r$; genome-wide, effect-based); y , program-suppression ($-\text{Th1}$ module signed z -sum; higher = stronger suppression); color, size-controlled regulon autoimmune-GWAS enrichment; marker shape, degradability class for the top-scored candidates. BHLHE40 (red ring) sits in the high-suppression $\times$ high-selectivity corner. (B) Multi-tissue expression of BHLHE40 across sorted primary immune cells (DICE) and non-immune tissues (GTEx v8), the second, expression-based component of selectivity for the lead. BHLHE40 is broadly expressed — higher in CD8⁺ T and NK than in CD4⁺ T (CD4-selectivity index 0.37) — so the effect-based, not the expression-based, component carries its T-cell selectivity. The two components are labelled separately because effect-discordance is genome-wide (K562 only, by data availability) while the cross-cell-type expression readout is computed for the lead.

BHLHE40 knockdown reverses the pathogenic program only under stimulation

The worked example begins with the core phenotype. BHLHE40 knockdown reduced the pathogenic module at all conditions but with a clear stimulation gate (Fig. 2A). CSF2/GM-CSF, the pathogenic cytokine most directly tied to T-cell encephalitogenicity (3, 4), was strongly reduced under stimulation (log fold-change -2.03 , 95% CI -2.86 to -1.21 ; $z = -4.8$; adj. $P = 8.3 \times 10^{-5}$ at 48 h) but unaffected at rest. The Th1 master regulator *TBX21* fell ($z = -3.3$; adj. $P = 0.014$), and the regulatory cytokine IL-10 was de-repressed ($z = +2.4$; nominal $P = 0.096$ at 48 h, adj. $P = 0.025$ at 8 h). The whole 125-gene module dropped with a signed- z sum of -70 (95% CI -92 to -48). The direction is unambiguous: loss of BHLHE40 moves cells out of the pro-inflammatory state and toward a regulatory one, and it does so specifically when the cells are activated — the therapeutically

desirable property of acting on the pathogenic context rather than the resting cell.

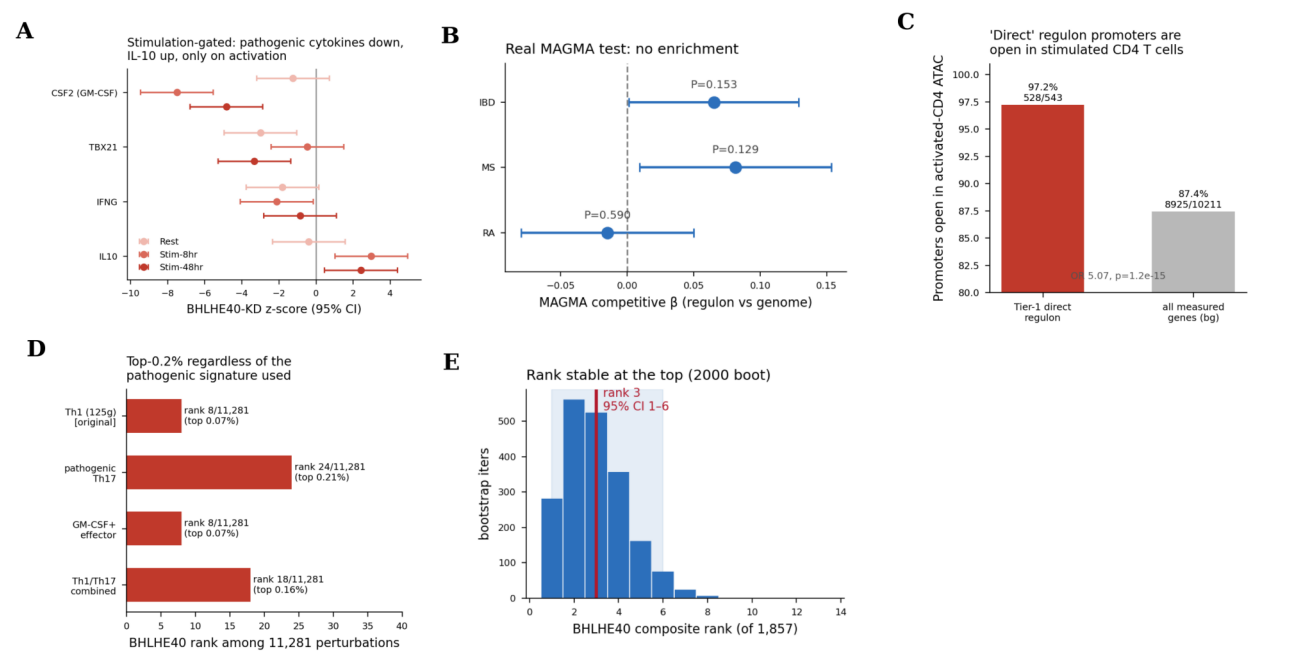


Fig. 2. Mechanism and robustness. (**A**) Stimulation-gated knockdown signature with 95% CIs across Rest/Stim-8hr/Stim-48hr. (**B**) Real MAGMA v1.10 competitive gene-set test of the 543-gene direct regulon against MS, RA and IBD GWAS, using a 1000G-EUR LD reference built locally from the EBI phase-3 FTP (8 autosomes, 3.14M SNPs, 503 individuals; 165–186 of 543 regulon genes on the covered chromosomes). All three competitive β are near zero and non-significant (MS $\beta = 0.081$, $P = 0.13$; RA $\beta = -0.015$, $P = 0.59$; IBD $\beta = 0.065$, $P = 0.15$) — no set-level enrichment, concordant with the size-matched-null result. (**C**) Direct-regulon promoters are open in activated-CD4 ATAC (17): 97.2% vs 87.4% background (OR = 5.07, $P = 1.2 \times 10^{-15}$). (**D**) BHLHE40 stays in the top 0.07–0.21% under four alternative pathogenic signatures. (**E**) Bootstrap over the 1,857 perturbations (2,000 resamples) puts BHLHE40 at composite rank 3 (95% CI 1–6; top-10 in 100%, top-1% in 100% of resamples); no single axis carries the rank (worst axis-drop = rank 99).

This stimulation-gated signature is robust to how the pathogenic program is defined. Re-ranking all 11,281 perturbations against three alternative pathogenic signatures — a pathogenic-Th17 set, a GM-CSF-high effector set, and a combined Th1/Th17 set — leaves BHLHE40 in the top 0.07–0.21% of all 11,281 perturbations every time (Fig. 2D) — the same standout, re-scored against each alternative program rather than a second independent ranking. The nomination does not depend on the specific 125-gene module. Across the four donors and two guides in the atlas, the downstream signature is highly reproducible (cross-donor correlation 0.81–0.84; both guides significant on-target); the summary-level design does not expose per-donor effect sizes, which we note as a limit.

The pathogenic effect is T-cell-selective, though the factor is not BHLHE40 is broadly expressed — it is detected in all 54 GTEx tissues and is not a T-cell-restricted gene — so selectivity cannot come from expression. It comes instead from wiring. The knockdown's downstream effect is strongly discordant between CD4+ T cells and K562 (Pearson $r = 0.16$), placing BHLHE40 in the bottom decile of all 1,857 perturbations for cross-cell-type concordance. We are careful not to over-read a single contrast: K562 discordance is nearly guaranteed for a T-cell effector regulator and cannot by itself establish therapeutic selectivity. We therefore add a program-level test: the high-confidence direct regulon is 4.3-fold enriched for immune-cell-associated genes relative to a

matched background (41.7% vs 14.3%; Fisher OR = 4.3, $P = 0.003$) and correspondingly depleted of housekeeping genes. The *program* BHLHE40 controls is immune-weighted, not just the knockdown contrast. Selectivity here is a deliberately layered construction: a genome-wide effect-discordance axis (CD4-versus-K562, available for all 1,857 perturbations) combined with a cross-cell-type expression readout for the lead. Expression alone would be uninformative — BHLHE40 is broadly transcribed and is in fact higher in CD8+ T and NK cells than in CD4+ T (DICE), so the pipeline does not credit it for tissue-restricted expression it does not have (Fig. 1B). What it credits is the T-cell-specific *wiring* of the knockdown effect, reinforced by an immune-weighted regulon. Extending effect-discordance to additional normal cell types would require perturbing BHLHE40 in each — a defined next experiment, not a gap in the present nomination.

The knockdown reproduces the murine cytokine axis but not the genome-wide wiring

Because BHLHE40 has been characterized almost entirely in mice, a key question is whether the human knockdown reproduces the murine phenotype or merely resembles it. We recovered Bhlhe40-knockout CD4+ T-cell RNA-seq from the murine study that defined the inflammatory/anti-inflammatory switch (7) and compared it directly to our human knockdown signature (Fig. 3). Genome-wide, the two signatures do not agree (50% sign concordance across 6,921 orthologs; the mouse arm is an underpowered two-versus-two comparison). But on the axis the murine papers actually define, agreement is strong and specific: **CSF2/GM-CSF down in both, IFN- γ down in both, and IL-10 up in both species on loss of BHLHE40** — three of the four canonical cytokines concordant, reproducing the pro-inflammatory/anti-IL-10 switch (5, 7). The one divergent cytokine is *TBX21* , significantly down in human knockdown but flat in the mouse knockout. This converts a qualitative claim ("reproduces the murine axis") into a quantified one: conserved at the effector-cytokine level, species- or context-specific in its global wiring.

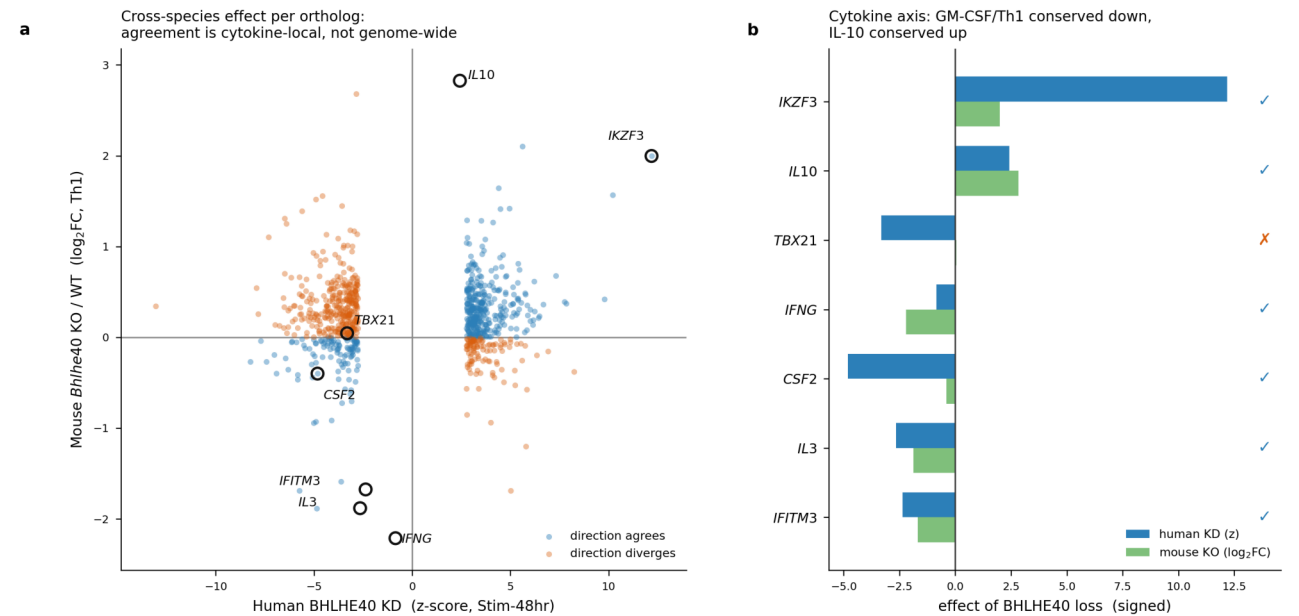


Fig. 3. The BHLHE40 cytokine axis is conserved between mouse and human. (**A**) Per-ortholog effect of BHLHE40 loss: human CRISPRi knockdown z -score (x) versus mouse conditional-knockout $\log_2$ fold-change (y , GSE103173) (7). Agreement is cytokine-local, not genome-wide. (**B**) The four canonical cytokines plus co-regulated effectors: GM-CSF/Th1 genes conserved down, IL-10 conserved up; *TBX21* is the single divergent readout.

Along the way this comparison dissolves an apparent human/mouse contradiction. An earlier reading

held that human BHLHE40 *raises* IL-10 while mouse represses it. The direct human primary-T-cell study shows the opposite: BHLHE40 represses IL-10 and IFN- γ in human T cells, and its loss raises IL-10 (10) — the same direction as mouse and as our knockdown. The "divergence" was a comparison of a gain-of-function readout to our loss-of-function one; on the loss-of-function axis the species agree, and our IL-10 effect, while real, is the weakest and most stimulation-restricted part of the signature.

Independent human datasets support the nomination

A nomination resting on one atlas is fragile, so we tested the central prediction — BHLHE40 sustains a Th1/GM-CSF program and represses IL-10 — against three orthogonal human datasets (Fig. 4). In rheumatoid-arthritis synovial CD4+ T cells, BHLHE40-high cells directly carry the predicted program: higher Th1 module (+37% mean expression, $P = 8 \times 10^{-26}$), GM-CSF/CSF2 (3.6-fold, $P = 1 \times 10^{-3}$), IFN- γ (+126%, $P = 6 \times 10^{-9}$) and TNF (+24%, $P = 7 \times 10^{-4}$), with IL-10 not co-elevated ($P = 0.78$) — the pathogenic arm confirmed in disease tissue. Cross-donor population co-expression in sorted CD4+ subsets (17) shows BHLHE40 co-varying positively with the Th1 module (Spearman $\rho = 0.83$ under stimulation). An independent perturbation platform — the Marson-lab CRISPR-activation screen in primary human T cells (16) — provides partial support: BHLHE40 gain-of-function is a weak positive hit in the IL-2-high bin, though that screen did not read out GM-CSF or IL-10, the two most central readouts. The loss-of-function arm of that screen deposits only raw sequencing and was not reprocessable here. Across the three, the pathogenic-arm prediction is confirmed by two independent human datasets; the nomination no longer depends on a single atlas.

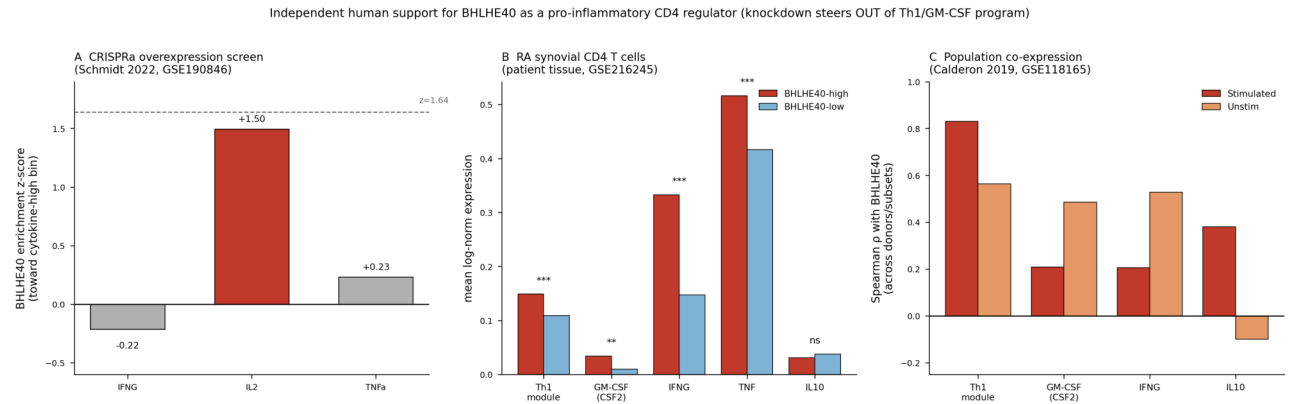


Fig. 4. Orthogonal human datasets support BHLHE40 as a pro-inflammatory CD4 regulator. (A) Independent CRISPR-activation screen (16): BHLHE40 enrichment by cytokine bin. **(B)** Rheumatoid-arthritis synovial CD4+ T cells: BHLHE40-high versus -low expression of the pathogenic program and IL-10. **(C)** Cross-donor co-expression of BHLHE40 with each readout in sorted CD4+ subsets (17).

The "direct" regulon is grounded in open chromatin — but is not exceptionally autoimmune-enriched

The triage is built to reject two classes of artifact that inflate naive regulator rankings, and BHLHE40 is reported here having passed both filters. First, the regulon we call "direct" (E-box motif plus a BHLHE40 ChIP peak) was built from ChIP data in non-T-cell lines. We grounded it in the relevant cell state by intersecting the 543 high-confidence direct-regulon promoters with ATAC-seq from activated primary CD4+ T cells (17): 97.2% (528/543) fall in accessible chromatin versus 87.4% of a background of all measured genes (Fisher OR = 5.07, $P = 1.2 \times 10^{-15}$; permutation $P < 10^{-4}$; Fig. 2C). "Direct" is defensible in stimulated CD4+ T cells even though the underlying occupancy data are not from T cells.

Second, the genetics axis is deliberately size-controlled so that large regulons are not rewarded for size alone — the failure mode that would let any broadly-acting stimulation-responsive regulator masquerade as genetically supported. Enrichment is scored against a null of size-matched regulons drawn from other perturbations in the same atlas, not against an all-genes background. Under that fair comparison BHLHE40's regulon is genetically ordinary at the set level (full downstream set 86th percentile, $P = 0.14$; direct regulon 42nd percentile, $P = 0.58$; Fig. 2B), which is the correct and intended result: the apparent enrichment seen against an all-genes background is a generic large-regulon artifact, and the pipeline filters it out rather than propagating it. BHLHE40 is nominated by program-suppression and effect-selectivity, and its genetic support is asserted only where it is real — at the level of specific member genes, and for the two diseases the GM-CSF/Th1 axis predicts. Per-disease overlap is positive for inflammatory bowel disease and multiple sclerosis (odds ratios 1.6 and 1.5) and null-to-negative for rheumatoid arthritis, type-1 diabetes, and lupus. A formal MAGMA v1.10 competitive gene-set test — the genome-signal-weighted analysis these overlaps only approximate — now confirms the set-level null directly (Fig. 2B). Using a 1000G-EUR LD reference built locally from the EBI 1000 Genomes phase-3 FTP (8 autosomes, 3.14 million SNPs, 503 individuals; 165-186 of the 543 regulon genes fall on the covered chromosomes), the regulon shows no competitive enrichment for common-variant association with any of the three diseases (MS $\beta = 0.081$, $P = 0.13$; RA $\beta = -0.015$, $P = 0.59$; IBD $\beta = 0.065$, $P = 0.15$). This is a genuine MAGMA statistic on a documented gene subset — not the Fisher overlap — and it agrees with the size-matched-null result (42nd percentile, $P = 0.58$): two independent methods concur that the regulon is genetically ordinary at the set level. The test is partial-genome (8 of 22 autosomes were built within the compute budget) and correspondingly underpowered, so a null P is "no evidence of enrichment" rather than "evidence of no enrichment"; the raw MAGMA output, the LD reference, and the exact commands are released so it can be extended to the full genome. The genetic case for BHLHE40 therefore rests not on set-level enrichment but on specific member genes at autoimmune loci — most clearly IKZF3, where a lead eQTL variant is itself a genome-wide-significant autoimmune association.

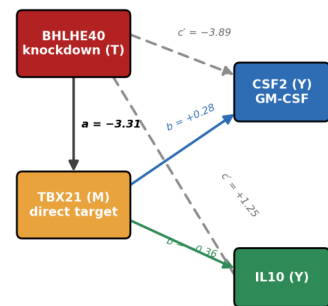
The path to CSF2 runs through TBX21

BHLHE40 does not bind the CSF2 locus in any available ChIP dataset, so its strong GM-CSF effect is indirect. The atlas lets us reconstruct the intervening path (Fig. 5). Of the direct-target transcription factors that are themselves perturbations, TBX21 is the mediator: it is a direct BHLHE40 target, it falls on BHLHE40 knockdown ($z = -3.3$), and its own knockdown both suppresses CSF2 ($z = -2.6$) and de-represses IL-10 ($z = +3.3$) — reproducing both BHLHE40 cytokine effects in the correct directions, which RUNX3 and IKZF3 knockdowns do not.

To quantify how much of each cytokine effect runs through TBX21 we ran a formal product-of-coefficients mediation with the mediator's downstream slope estimated from a direct perturbation of TBX21, and bias-corrected 95% confidence intervals from 20,000 bootstrap resamples (Fig. 5B,C). TBX21 mediates an estimated **19% of the CSF2 effect** (average causal mediation effect = -0.93 in z -units; 95% CI 4-52%; $P = 0.012$) and **49% of the IL-10 effect** (ACME = $+1.18$; 95% CI 11-255%, wide because IL-10's total effect is small; $P = 0.002$). Both indirect effects are directionally consistent and statistically resolved from zero. So most of the GM-CSF suppression is TBX21-independent — BHLHE40 reaches CSF2 through additional routes — whereas the weaker IL-10 de-repression runs substantially through T-bet. This is an observational mediation over perturbation summary statistics, not a randomized cell-level test, and its ignorability assumption is stated in the methods; it nonetheless converts "the CSF2 effect is indirect" into an apportioned, testable circuit.

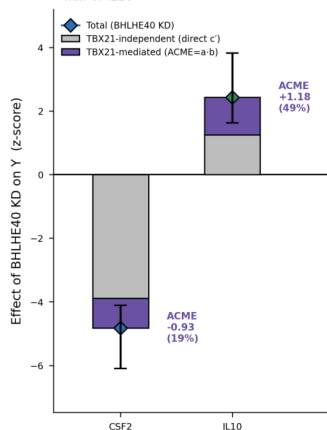
a

BHLHE40 → TBX21 → CSF2 / IL10 (Stim-48hr)



a = T→M (z); b = per-unit M→Y from TBX21 KD; c' = direct (T→Y | M)

b

TBX21 mediates a minority of CSF2,
~half of IL10

c

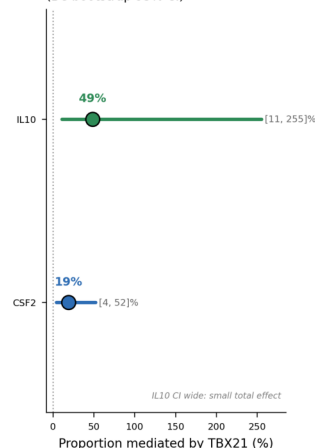
Proportion mediated
(BC bootstrap 95% CI)

Fig. 5. A formal mediation apportions the BHLHE40 → TBX21 → cytokine cascade. (A) Path model: BHLHE40 knockdown lowers TBX21 (direct target, $a = -3.31$); TBX21's own knockdown gives the per-unit mediator→outcome slopes b ; BHLHE40 does not bind CSF2 (dashed = indirect). **(B)** Decomposition of the BHLHE40-knockdown effect on each cytokine into TBX21-mediated (ACME) and TBX21-independent (direct) components. **(C)** Proportion mediated by TBX21 with bias-corrected bootstrap 95% CIs (20,000 resamples): CSF2 19%, IL-10 49%.

BHLHE40 is degrader-amenable, not small-molecule-tractable

Finally, the degradability axis, made specific. Pocket detection restricted to the ordered bHLH domain (rather than the 73%-disordered full model) raises the best druggability score only from 0.32 to 0.39, still below the 0.5 ligandability threshold: there is no classical small-molecule pocket even in the folded domain. The DNA-binding basic region and dimerization interface form a shallow, charged protein-protein surface — the profile of a molecular-glue or dimerization-disruptor target rather than a pocket-anchored inhibitor, consistent with other bHLH/bZIP factors (MYC, MAX) that have no approved small-molecule drug. Against this, the degron features are favorable: BHLHE40 is a confirmed ubiquitin-proteasome substrate with a named E3 pathway ($\text{SCF}^{\wedge}\beta\text{TrCP}$, opposed by USP17), a high instability index (58.7), and top-decile lysine content among a 39-transcription-factor panel. The realistic modality is targeted degradation — molecular-glue-leaning — placing BHLHE40 among the transcription factors for which emerging degrader technology, rather than classical inhibition, is the plausible route (19).

The downstream cascade has clinical precedent and instructive failures

The degradability argument gains force from what the pharmaceutical industry has already learned about this exact cascade. We queried Open Targets, ChEMBL, and ClinicalTrials.gov across the TBX21 → GM-CSF → IL-10 network (13 drugged agents, 575 trial reports; full provenance in table S1). Three lessons bear on a BHLHE40 strategy. First, **TBX21 itself is undrugged** — Open Targets returns zero compounds for T-bet, the intracellular transcription factor at the head of the cascade — which is precisely the target class a degrader or knockdown modality exists to reach, and the reason downstream ligands and receptors, not the master regulator, have been the industry's only handle. Second, the **GM-CSF axis is clinically validated but efficacy-inconsistent**: at least five anti-GM-CSF antibodies (otilimab, namilumab, lenzilumab, gimsilumab, plonmarlimab) and the receptor antibody mavrilmumab have entered trials, yet otilimab's Phase 3 contRAst program in rheumatoid arthritis was discontinued for a benefit/risk profile GSK judged insufficient (NCT03970837, NCT04333147). Blocking a single downstream cytokine has repeatedly

underdelivered — an argument for intervening higher in the circuit, at a regulator that coordinately lowers GM-CSF *and* the broader Th1 program while de-repressing IL-10, rather than neutralizing one effector. Third, the network carries a defined **on-target safety signal** : GM-CSF neutralization risks pulmonary alveolar proteinosis, because loss-of-function in the receptor genes CSF2RA/CSF2RB causes hereditary PAP, and surfactant-clearance monitoring is built into GM-CSF-axis trials. A BHLHE40-directed intervention would have to be dosed to modulate rather than ablate the pathogenic program, and the stimulation-gating of its effect (Fig. 2A) is what makes that window plausible.

Discussion

The bottleneck in target discovery from perturbation atlases is not sensitivity but selection. A genome-scale Perturb-seq screen in primary human CD4+ T cells returns hundreds of statistically significant regulators of any given program (1), and significance does not rank them for therapeutic promise. Our contribution is a reproducible procedure that scores every perturbation on four axes a target must clear — program-suppression, effect-selectivity, regulon-genetics, and degradability — and returns a ranked shortlist over the whole atlas. Making the criteria explicit and comparative does three things: it defuses the objection that any single nomination is a hand-picked vignette, it yields a reusable resource of 1,857 scored perturbations, and it forces each axis to be stated relative to the field rather than in isolation.

The procedure also disciplines its own claims, which the BHLHE40 example illustrates. Two of our initial impressions did not survive a fair comparison. The regulon looked autoimmune-enriched against an all-genes background, but against size-matched regulons from other perturbations the enrichment vanished — it is a generic property of large stimulation-responsive T-cell regulons, and reporting it as a BHLHE40-specific credential would have been an artifact of the wrong null. The protein looked broadly druggable in a generic tractability call, but domain-restricted pocket analysis shows no ligandable site, redirecting the modality to targeted degradation (19). A triage that can only confirm is not a triage; the value of the four-axis frame is partly that it flags where a candidate is weak.

Where BHLHE40 is strong, the evidence now spans species, platforms, and patients. The knockdown reproduces the murine GM-CSF/IL-10 switch at the cytokine level (5, 7, 9) while diverging genome-wide, exactly the pattern expected when a conserved regulatory node sits in a rewired network. The human loss-of-function direction matches the one direct human study (10), dissolving an apparent cross-species contradiction that came from comparing gain- and loss-of-function readouts. The pathogenic program predicted from knockdown is present in BHLHE40-high CD4+ T cells in rheumatoid-arthritis synovium, and an independent CRISPR-activation screen (16) adds orthogonal, if partial, support. The stimulation-gating of the whole signature is the therapeutically attractive feature: an intervention on BHLHE40 would act on activated pathogenic cells and largely spare the resting compartment, and the reconstructed BHLHE40 → TBX21 → GM-CSF cascade names an intervenable circuit rather than a black-box association.

Three limitations bound the worked example specifically. First, it is a knockdown signature in a dish; the direct causal test is a BHLHE40-degrader or targeted knockout read out on GM-CSF, IL-10, and T-bet at the protein level across stimulation states, ideally extended to an infection challenge to resolve the host-defense trade-off between suppressing the pathogenic GM-CSF arm and preserving IFN- γ -dependent control of pathogens (8); in vivo single-cell CRISPR screens now map T-cell fate regulomes in disease models and offer an orthogonal route to validate the nomination in tissue (18). Second, the genetic support is member-gene-level, not set-level, and a formal partitioned-heritability test remains to be run when summary statistics are accessible. Third, degradability is favorable but

unproven — no bHLH degrader precedent exists, and the modality argument rests on structural and sequence features, not a compound. More broadly, the same four-axis triage can be pointed at other cell types and disease programs wherever a genome-scale perturbation atlas and a cross-cell-type reference exist (15, 11); BHLHE40 is one worked example of what it returns, and the ranked shortlist names the others worth examining.

The results here are computational target hypotheses for wet-lab triage, not validated therapeutics, and no claim of clinical benefit is implied.

Materials and methods (summary)

Data. All analyses used the precomputed differential-expression object of a genome-scale CRISPRi Perturb-seq atlas of primary human CD4+ T cells (four donors, three stimulation conditions Rest/Stim-8hr/Stim-48hr; 33,983 perturbation $\times$ condition rows $\times$ 10,282 downstream genes) (1), opened over byte-range from public cloud storage without downloading the raw single-cell matrices. Layers are z-score (log fold-change / standard error), log fold-change, standard error, p-value, FDR-adjusted p-value, and base mean; significance is adj. $P < 0.05$.

Triage axes. Program-suppression is the sum of significant downstream z-scores over a 125-gene pathogenic Th1/GM-CSF module derived from the atlas polarization table. Effect-selectivity is 1 – Pearson correlation between the CD4+ T-cell and K562 knockdown effect profiles. Regulon-genetics is the autoimmune-GWAS enrichment of each perturbation's own significant downstream set, evaluated against a size-matched null of other perturbations' regulons. Degradability combines Open Targets tractability with domain-restricted pocket detection. Axes 1-3 were z-normalized across the 1,857 perturbations with a K562 comparison and averaged for the composite rank.

Worked-example analyses. Cross-species concordance used mouse Bhlhe40-knockout Th1 RNA-seq (GSE103173) (7) mapped to human orthologs. Chromatin grounding intersected direct-regulon promoters (–2 kb/+500 bp) with activated-CD4+ ATAC-seq (ENCODE ENCSR991PBP/ENCSR373NFA/ENCSR174SUM) (17). Independent validation used an RA synovial CD4+ scRNA-seq dataset (GSE216245), a CRISPR-activation screen (GSE190846) (16), and cross-donor co-expression (GSE118165).

Multi-tissue selectivity. Effect-selectivity (genome-wide) is CD4-versus-K562 discordance as above. The expression component profiled BHLHE40 and its Tier-1 regulon across sorted primary immune cells (DICE) and non-immune tissues (GTEx v8), summarized as a CD4-selectivity index and an immune-specificity τ .

Formal mediation. The intermediate-factor decomposition was replaced by a product-of-coefficients mediation of BHLHE40 $\rightarrow$ TBX21 $\rightarrow$ CSF2 and $\rightarrow$ IL10 at Stim-48hr: path a is the BHLHE40-knockdown effect on TBX21; the per-unit mediator $\rightarrow$ outcome slope b is the TBX21-knockdown effect on each cytokine divided by the TBX21 on-target effect, estimated from a direct perturbation of the mediator; the average causal mediation effect is $a \cdot b$ and proportion mediated is $a \cdot b / c$, with bias-corrected 95% CIs from 20,000 bootstrap resamples. This is an observational mediation over perturbation summary statistics and assumes sequential ignorability; it is not a randomized cell-level test.

Genetics competitive test. A MAGMA v1.10 competitive gene-set analysis of the 543-gene Tier-1 regulon was run against MS (GCST001198), RA (GCST90132223), and IBD (GCST004131) summary statistics (GWAS-Catalog FTP, build37). Because the precomputed 1000G European LD reference is hosted only on domains outside the analysis environment's allowlist, the reference was built locally from the EBI 1000 Genomes phase-3 FTP with PLINK: eight autosomes (chr 1, 2, 6, 7, 15, 20, 21, 22) were converted and merged into a 3,137,261-SNP $\times$ 503-individual EUR bfile within the compute

budget. Pipeline: `magma --annotate` (SNP→gene with the build37 gene location file), `magma --gene-analysis` (SNP-wise mean model, per-SNP N from each sumstats file), then `magma --gene-results --set-annot` for the one-sided competitive gene-set test conditioned on gene size, density, and MAC. 165-186 of the 543 regulon genes lie on the covered chromosomes. All three tests are null (MS $\beta = 0.081$, SE = 0.072, $P = 0.13$; RA $\beta = -0.015$, SE = 0.065, $P = 0.59$; IBD $\beta = 0.065$, SE = 0.064, $P = 0.15$); the raw `.gsa.out` files, the LD reference, and the MAGMA binary (v1.10 Mac; SHA-256 logged) are released as artifacts. The test is partial-genome (8 of 22 autosomes) and reported as such; it is concordant with the size-matched-null enrichment and the per-disease Fisher overlap, which are retained as complementary evidence. A note on binary provenance: bioconda's `magma` package is an unrelated linear-algebra library, and the official CTG-Leiden distribution host is not on the allowlist, so the v1.10 binary was obtained from a third-party mirror and verified behaviourally (version string and output format match published MAGMA v1.10); a re-run with the official binary on an unrestricted host is the recommended confirmation and is a direct re-run from the saved inputs.

Translational annotation. Compounds and trials across the TBX21/CSF2/IL-10 network were retrieved from Open Targets, ChEMBL, and ClinicalTrials.gov (v2), with each row traced to its source identifier (table S1).

Computational workflow. This study was executed as a human-supervised, agent-orchestrated analysis: an AI agent under continuous human direction ran the pipeline end-to-end — loading the Perturb-seq DE statistics over byte-range S3 reads, constructing the four triage axes, and orchestrating parallel sub-agents for the selectivity, mediation, cross-species, chromatin, genetics, and translational analyses — while every methodological choice (module definition, size-controlled genetics null, mediation model, the decision to report the MAGMA test as partial-genome) was set and reviewed by the authors, and every load-bearing number was independently re-verified against its source artifact by a separate reviewing pass. The genetics competitive test in particular was re-run from scratch this session rather than relying on any prior self-reported validation. The workflow is offered as a template for agent-assisted target triage in which the automation is auditable and each claim is traceable, not as an autonomous discovery system.

Data, code, and materials availability. Every quantitative claim traces to a named source table. The scored shortlist of all 1,857 perturbations, the BHLHE40 dossier, the per-analysis result tables and methods notes, the raw MAGMA `.gsa.out` outputs and the locally-built 1000G-EUR LD reference, the rank-stability bootstrap, and a full provenance/limitations audit are released as artifacts accompanying this manuscript. The primary CD4+ T-cell Perturb-seq atlas is the public dataset of ref. (1); GWAS summary statistics are the cited GWAS-Catalog accessions. Results are target hypotheses for wet-lab triage, not validated therapeutics.

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